

Credit Name: CSE2920 - CSE Project C

Assignment Name: TemperatureAndHumidity

How has your program changed from planning to coding to now? Please explain?

PLANNING:

I plan to modify the given code so that the program only prints humidity above 30%, and temperatures above 21 degrees C. If humidity is under 30%, the program will print "Humidity is low", and if the temperature is below 21 degrees C, the program will print "Room is too cold."

CODING:

1.

```
//Add Phidgets Library | You added a file called phidget22 when configuring
import com.phidget22.*;

public class Temperatureandhumidity {
    public static void main(String[] args) throws Exception{

        //Create | Here you've created a HumiditySensor and a TemperatureSensor
        HumiditySensor humiditySensor = new HumiditySensor();
        TemperatureSensor temperatureSensor = new TemperatureSensor();

        //Open | Open establishes a connection between your object and the Phidgets box
        humiditySensor.open(1000);
        temperatureSensor.open(1000);
    }
}
```

Phidgets library is imported

The temperatureSensor and humiditySensor objects are created to interact with the physical phidgets box.

Opened both objects, establishing a connection to the physical Phidgets box.

2.

```
//Use your Phidgets | This code will print humidity and temperature read by the sensor
while(true){

    if (humiditySensor.getHumidity()/100 < 0.3) {
        System.out.println("Humidity is low");
        Thread.sleep(500);
    }
    else {
        System.out.println("Humidity: " + humiditySensor.getHumidity() + " %RH ");
        Thread.sleep(150);
    }

    if (temperatureSensor.getTemperature() > 21) {
        System.out.println(" Temperature: " + temperatureSensor.getTemperature() + " °C");
        Thread.sleep(150);
    }
    else {
        System.out.println("Room is too cold.");
        Thread.sleep(500);
    }
}
```

While loop is used so that the following code repeats.

If humidity divided by 100 is less than 0.3 (if humidity is less than 30%), the program prints "Humidity is low". Sleep timer of 500ms since there is no humidity output, no need to excessively print "Humidity is low".

Else, the program prints the humidity with a sleep timer of 150ms so that the humidity is printed in a timely manner.

If the temperature is greater than 21, the program outputs the temperature in Celsius with a sleep timer of 150ms.

Else, the program prints "Room is too cold." Sleep timer of 500ms.

End Of Program!